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DUAL-POLARIZED ANTENNA ELEMENT, ANTENNA APPARATUS, AND BASE STATION SYSTEM

Abstract

A dual-polarized antenna element including a radiator having a first feeding part and a second feeding part, and having a through hole between the first feeding part and the second feeding part, a first feeding structure coupled to the first feeding part, a second feeding structure coupled to the second feeding part, a first balancer that passes through the through hole and that has a first coupling part and a first ground part, the first coupling part and ground part being respectively located on two sides of the radiator, and the first coupling part being coupled to the radiator, and a second balancer that passes through the through hole, and that has a second coupling part and a second ground part, the second coupling part and ground part being respectively located on the two sides of the radiator, and the second coupling part being coupled to the radiator.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2023/117373, filed on Sep. 7, 2023, which claims priority to Chinese Patent Application No. 202211397101.8, filed on Nov. 9, 2022. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] This application relates to the field of communication technologies, and in particular, to a dual-polarized antenna element, an antenna apparatus, and a base station system.

BACKGROUND

[0003] For a conventional dual-polarized antenna, radiation arms of the dual-polarized antenna are independent of each other, and the radiation arms and a feeding structure are independent of each other. Consequently, manufacturing, processing, and assembly of the radiation arms and the feeding structure are complex, costs are high, and integrity of the antenna is poor.

SUMMARY

[0004] An objective of this application is to provide a dual-polarized antenna element, an antenna apparatus, and a base station system, to resolve problems in the conventional technologies that manufacturing, processing, and assembly of a dual-polarized antenna are complex, costs are high, and integrity is poor.

[0005] A first aspect of this application provides a dual-polarized antenna element, including a radiator, where a first feeding part and a second feeding part are integrally formed on the radiator, the radiator is provided with a through hole, and the through hole is located between the first feeding part and the second feeding part, a first feeding structure, coupled to or directly connected to the first feeding part, a second feeding structure, coupled to or directly connected to the second feeding part, a first balancer, where the first balancer passes through the through hole, the first balancer includes a first coupling part and a first ground part, the first coupling part and the first ground part are respectively located on two sides of the radiator, and the first coupling part is coupled to the radiator, and a second balancer, where the second balancer passes through the through hole, the second balancer includes a second coupling part and a second ground part, the second coupling part and the second ground part are respectively located on the two sides of the radiator, and the second coupling part is coupled to the radiator.

[0006] The first feeding part and the second feeding part are integrally formed on the radiator, so that the radiator, the first feeding part, and the second feeding part form a mechanical part. In other words, the radiator, the first feeding part, and the second feeding part are integrally formed by using an independent sheet metal part, so that a structure is simplified, and integrity of an antenna is improved. The first feeding part and the second feeding part are located on two sides of the through hole, so that isolation between parts near the first feeding part and the second feeding part can be improved, thereby improving radiation performance.

[0007] Through the provision of the first balancer and the second balancer, an electric field of the radiator may be more even, so that isolation between two parts whose polarization directions are

different on the radiator can meet a requirement on isolation between antennas. Therefore, a stable dual-polarized characteristic can be formed on one radiator, and the dual-polarized antenna element can achieve a stable directivity pattern. In addition, a size of the antenna element can be reduced, miniaturization can be implemented, and space occupation can be reduced, thereby facilitating integration into a base station antenna.

[0008] In a possible design, the first feeding part includes one first feeding bent section, and the first feeding bent section is connected to the first feeding structure to generate a first electromagnetic wave signal in a first polarization direction. The second feeding part includes one second feeding bent section, and the second feeding bent section is connected to the second feeding structure to generate a second electromagnetic wave signal in a second polarization direction.

[0009] In a possible design, the first feeding part includes two first feeding bent sections, and the two first feeding bent sections are disposed in a centrally symmetric manner to generate a first electromagnetic wave signal in a first polarization direction. The second feeding part includes two second feeding bent sections, and the two second feeding bent sections are disposed in a centrally symmetric manner to generate a second electromagnetic wave signal in a second polarization direction. In this way, radiation performance of the antenna can be further improved.

[0010] In a possible design, in a thickness direction of the radiator, projection of the first coupling part and projection of at least a part of the radiator overlap, and projection of the second coupling part and projection of at least a part of the radiator overlap. In this way, the first coupling part can separately correspond to parts that are on the radiator and that are located on two sides of a first gap, thereby improving an effect of current coupling between the first coupling part and the radiator, and helping make an electric field near the first gap more even, to obtain a stable directivity pattern. Similarly, the second coupling part also has a same effect as the first coupling part, and details are not described herein again. Therefore, the first coupling part and the second coupling part can ensure evenness of an electric field of the dual-polarized antenna element, and ensure stability of the directivity pattern.

[0011] In a possible design, the first balancer includes a first connection section. The first connection section passes through the through hole. The first coupling part and the first ground part are respectively connected to two ends of the first connection section. An angle between the first connection section and a radiation surface of the radiator ranges from 80° to 100° . In this way, the first connection section can be made nearly perpendicular to the radiation surface of the radiator, in other words, a direction of a current on the first connection section is nearly perpendicular to a direction of the electric field of the radiator, so that the current on the first connection section can be prevented from being coupled with a current on the radiator. Therefore, the antenna can achieve excellent radiation performance and a stable directivity pattern.

[0012] In a possible design, the second balancer includes a second connection section. The second connection section passes through the through hole. The second coupling part and the second ground part are respectively connected to two ends of the second connection section. An angle between the second connection section and the radiation surface of the radiator ranges from 80° to 100° . A bending direction of the second coupling part and a bending direction of the first coupling part are opposite. A function and an arrangement effect of the second connection section are the same as those of the first connection section, and details are not described herein again. The bending direction of the first coupling part and the bending direction of the second coupling part are away from each other, so that the first coupling part and the second coupling part can be separately coupled to the radiator near the first gap and a second gap, thereby improving an electric field balance effect and helping obtain a stable directivity pattern.

[0013] In a possible design, an angle between the first coupling part and the radiation surface of the radiator and an angle between the second coupling part and the radiation surface of the radiator range from 0° to 10° . In this way, the first coupling part and the second coupling part can have large areas and can be coupled to the radiator, thereby improving an effect of even distribution of

the electric field.

[0014] In a possible design, the first coupling part and the second coupling part are separately parallel to the radiation surface of the radiator.

[0015] In a possible design, a distance between the first coupling part and the radiation surface of the radiator and a distance between the second coupling part and the radiation surface of the radiator are less than 0.1 times a wavelength corresponding to a maximum operating frequency. In this way, a large bandwidth can be obtained, and it is ensured that the dual-polarized antenna element achieves good radiation performance and a stable directivity pattern.

[0016] In a possible design, a maximum distance between the first coupling part and the first ground part ranges from 0.25 times to 0.75 times a wavelength corresponding to a maximum operating frequency, and a maximum distance between the second coupling part and the second ground part ranges from 0.25 times to 0.75 times the wavelength corresponding to the maximum operating frequency. Therefore, the dual-polarized antenna element can achieve a constant directivity pattern, and has good radiation performance and isolation performance.

[0017] In a possible design, an angle between the first feeding part and the radiation surface of the radiator and an angle between the second feeding part and the radiation surface of the radiator range from 60° to 120° . In this way, the first feeding part and the second feeding part can be separately nearly perpendicular to the radiation surface of the radiator. In other words, a direction of a current on the first feeding part and a direction of a current on the second feeding part are nearly perpendicular to the direction of the electric field of the radiator, so that the current on the first feeding part and the current on the second feeding part can be prevented from being coupled with the current on the radiator before arriving at the radiator. Therefore, the antenna can achieve the excellent radiation performance and the stable directivity pattern.

[0018] In a possible design, the dual-polarized antenna element further includes a reflection panel. The reflection panel is located on a side that is of the radiator and that is away from the radiation surface of the radiator. The first ground part and the second ground part are separately coupled to or directly connected to the reflection panel.

[0019] The reflection panel can reflect an electromagnetic wave radiated by a surface of a side that is of the radiator and that is away from the radiation surface, so that a radiation direction of the reflected electromagnetic wave is consistent with a radiation direction of the radiation surface, thereby enhancing electromagnetic wave energy and improving the radiation performance of the antenna.

[0020] In a possible design, a plurality of dual-polarized antenna elements share one reflection panel. In this way, integrity of the plurality of dual-polarized antenna elements can be improved, and radiation performance of an antenna apparatus having the plurality of dual-polarized antenna elements can be improved.

[0021] In a possible design, the first ground part and the second ground part are connected to form an integrated structure. In this way, grounding consistency between the first balancer and the second balancer can be ensured. The first balancer and the second balancer may alternatively be designed as one part, facilitating production and manufacturing, and facilitating assembly of the balancer in the dual-polarized antenna element.

[0022] In a possible design, the dual-polarized antenna element further includes a third balancer and a fourth balancer. Structures of the third balancer and the fourth balancer are the same as structures of the first balancer and the second balancer. The third balancer and the fourth balancer both pass through the through hole. The third balancer is centrally symmetrical to the first balancer. The fourth balancer is centrally symmetrical to the second balancer. Through the provision of the four balancers, the dual-polarized antenna element can obtain a better polarization effect.

[0023] In a possible design, the radiator is a metal plate.

[0024] In a possible design, the first feeding part and the second feeding part are integrally formed on the radiator by using a stamping and bending process.

[0025] A second aspect of this application further provides an antenna apparatus, including the dual-polarized antenna element according to the first aspect of this application.

[0026] A third aspect of this application further provides a base station system, including the antenna apparatus according to the second aspect of this application.

[0027] It should be understood that the foregoing general descriptions and the following detailed descriptions are merely examples, and are not intended to limit this application.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. 1 is a diagram of a structure of a base station system in some embodiments according to this application;

[0029] FIG. 2 is a diagram of a structure of an antenna apparatus in some embodiments according to this application;

[0030] FIG. 3 is a diagram of a structure of a dual-polarized antenna element from a perspective according to an embodiment of this application;

[0031] FIG. 4 is a diagram of a structure of a dual-polarized antenna element from another perspective according to an embodiment of this application;

[0032] FIG. 5 is a diagram of a structure of a dual-polarized antenna element from still another perspective according to an embodiment of this application;

[0033] FIG. 6 is a diagram of a structure of a radiator according to an embodiment of this application;

[0034] FIG. 7 is a diagram of a structure of a radiator according to another embodiment of this application;

[0035] FIG. 8 is a diagram of a structure of a dual-polarized antenna element from a perspective according to another embodiment of this application;

[0036] FIG. 9 is a diagram of a structure of a dual-polarized antenna element from another perspective according to another embodiment of this application;

[0037] FIG. 10 is a diagram of a structure of a dual-polarized antenna element from another perspective according to another embodiment of this application;

[0038] FIG. 11 is a diagram of a structure of a first balancer and a second balancer in a dual-polarized antenna element according to another embodiment of this application;

[0039] FIG. 12 is a top view of a dual-polarized antenna element according to another embodiment of this application;

[0040] FIG. 13 is a side view of a dual-polarized antenna element according to another embodiment of this application;

[0041] FIG. 14 is a diagram of a structure of a dual-polarized antenna element from a perspective according to still another embodiment of this application;

[0042] FIG. 15 is a diagram of a structure of an antenna array according to an embodiment of this application;

[0043] FIG. 16 is a diagram of a structure of fitting of a first balancer, a second balancer, a third balancer, and a fourth balancer;

[0044] FIG. 17 is a diagram of a structure of a dual-polarized antenna element having four balancers according to an embodiment of this application; and

[0045] FIG. 18 is a diagram of a structure of a dual-polarized antenna element having four balancers according to another embodiment of this application.

REFERENCE NUMERALS

[0046] **100**—Antenna apparatus; [0047] **101**—Antenna array; [0048] **102**—Feeding network;

[0049] **103**—Phase shifter; [0050] **104**—Transmission network; [0051] **105**—Combiner; [0052]

106—Antenna connector; [0053] **107**—Radome; [0054] **200**—Antenna adjustment support; [0055] **300**—Fastening pole; [0056] **400**—Connector sealing member; [0057] **500**—Ground apparatus; [0058] **10**—Dual-polarized antenna element; [0059] **1**—Radiator; [0060] **11**—First feeding part; [0061] **111**—First feeding bent section; [0062] **12**—Second feeding part; [0063] **121**—Second feeding bent section; [0064] **13**—Through hole; [0065] **14**—First gap; [0066] **15**—Second gap; [0067] **16**—Gap; [0068] **17**—Radiation surface; [0069] **2**—First feeding structure; [0070] **3**—Second feeding structure; [0071] **4**—First balancer; [0072] **41**—First coupling part; [0073] **42**—First ground part; [0074] **43**—First connection section; [0075] **5**—Second balancer; [0076] **51**—Second coupling part; [0077] **52**—Second ground part; [0078] **53**—Second connection section; [0079] **6**—Third balancer; [0080] **7**—Fourth balancer; [0081] **8**—Reflection panel.

[0082] The accompanying drawings herein are incorporated into this specification and constitute a part of this specification, to show embodiments in accordance with this application, and are used together with this specification to explain the principle of this application.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0083] To better understand technical solutions of this application, the following describes embodiments of this application in detail with reference to the accompanying drawings. It should be understood that the specific embodiments described herein are merely used to explain this application but are not intended to limit this application.

[0084] In descriptions of this application, unless otherwise specified and limited, the terms “first” and “second” are merely intended for a purpose of description, and should not be understood as an indication or implication of relative importance. Unless otherwise specified or stated, the term “a plurality of” means two or more than two. The terms “connection”, “fastening”, and the like should be understood in a broad sense. For example, the “connection” may be a fixed connection, or may be a detachable connection, an integrated connection, or an electrical connection, or may be a direct connection, or may be an indirect connection by using an intermediate medium. A person of ordinary skill in the art may understand specific meanings of the foregoing terms in this application according to specific situations.

[0085] With rapid development of 5G communication technologies, it is required that a base station antenna can implement higher performance at limited costs. In this application scenario, a design of a radiation unit in the base station antenna also faces a higher challenge. For a conventional dual-polarized antenna, radiation arms of the dual-polarized antenna are independent of each other, and the radiation arms and a feeding structure are independent of each other. Consequently, manufacturing, processing, and assembly of the radiation arms and the feeding structure are complex, costs are high, and integrity of the antenna is poor.

[0086] As shown in FIG. 1 and FIG. 2, an embodiment provides a dual-polarized antenna element **10**, an antenna apparatus **100**, and a base station system. The base station system, the antenna apparatus **100**, and the dual-polarized antenna element **10** may be used in fields such as radar, broadcast, and communication. As shown in FIG. 1, the base station system includes the antenna apparatus **100**, an antenna adjustment support **200**, a fastening pole **300**, a connector sealing member **400**, a ground apparatus **500**, and the like. The base station system is an interface device for wireless communication and can exchange information with a communication terminal in a region in which the base station system is located.

[0087] As shown in FIG. 2, the antenna apparatus **100** includes an antenna array **101**, a phase shifter **103**, a transmission network **104** or a calibration network, a combiner **105** or a waver, and a radome **107**. The antenna array **101** receives or transmits a radio frequency signal by using a feeding network **102** formed by the phase shifter **103**, the transmission network **104**, and the combiner **105**. The feeding network **102** can feed the radio frequency signal to the antenna array **101** based on a specific amplitude and phase, or send a radio signal received by the antenna array **101** to a signal processing unit of the base station system through an antenna connector **106** based on a specific amplitude and phase. The radome **107** may protect an internal component from risks

such as electromagnetic interference in an external environment and damage by an external foreign object.

[0088] As shown in FIG. 3 to FIG. 5, the dual-polarized antenna element **10** includes a radiator **1**, a first feeding structure **2**, and a second feeding structure **3**.

[0089] The radiator **1** is an apparatus configured to receive/send electromagnetic waves. In some cases, the radiator converts guided wave energy from a transmitter into a radio wave, or converts a radio wave into guided wave energy, to radiate and receive the radio wave. Modulated high-frequency current energy (or guided wave energy) generated by the transmitter is transmitted to a transmitting radiator through a feeder. The radiator converts the energy into specific polarized electromagnetic wave energy and radiates the energy in a required direction. A receiving radiator converts specific polarized electromagnetic wave energy from a specific direction of space into modulated high-frequency current energy, and transmits the modulated high-frequency current energy to an input end of a receiver through a feeder.

[0090] In this embodiment, as shown in FIG. 6, a first feeding part **11** and a second feeding part **12** are integrally formed on the radiator **1**, and the radiator **1** is provided with a through hole **13**. The first feeding part **11** and the second feeding part **12** may be configured to feed power to a corresponding part of the radiator **1**, so that the radiator **1** generates $\pm 45^\circ$ dual-polarized electromagnetic waves.

[0091] The first feeding structure **2** is coupled to or directly connected to the first feeding part **11**. The second feeding structure **3** is coupled to or directly connected to the second feeding part **12**. The first feeding structure **2** and the second feeding structure **3** may feed power to the radiator **1** through the first feeding part **11** and the second feeding part **12** respectively. The first feeding structure **2** and the second feeding structure **3** may be conducting-wire transmission lines, coaxial transmission lines, waveguides, microstrips, or the like. In this embodiment, the first feeding structure **2** and the second feeding structure **3** are preferably microstrips. The coupling connection is a non-direct contact connection. To be specific, the first feeding structure **2** is not in contact with the first feeding part **11**, and the second feeding structure **3** is not in contact with the second feeding part **12**, to indirectly feed power to the radiator **1** in a coupling manner. The foregoing direct connection is a manner of implementing feeding through direct contact. In addition, the first feeding part **11** and the second feeding part **12** are integrally formed on the radiator **1**, so that the radiator **1**, the first feeding part **11**, and the second feeding part **12** form a mechanical part, thereby simplifying a structure of the dual-polarized antenna element.

[0092] Specifically, as shown in FIG. 6, the first feeding part **11** and the second feeding part **12** are integrally formed on the radiator **1** by using a stamping and bending process. The radiator **1** may be a sheet metal part, for example, a sheet metal plate or a metal sheet with a specific thickness. During molding, stamping may be performed at a specified position on the radiator **1**. A stamped part is the first feeding part **11** or the second feeding part **12**. After the stamping is performed, parts of the first feeding part **11** and the second feeding part **12** are still connected to the radiator **1** and are not completely crushed, and the connected parts may be bent by using a bending process, a specific angle is formed between the first feeding part **11** and a radiation surface **17** of the radiator **1** and a specific angle is formed between the second feeding part **12** and the radiation surface **17** of the radiator **1**, to complete manufacturing of the first feeding part **11** and the second feeding part **12**. Therefore, in this embodiment, the radiator **1**, the first feeding part **11**, and the second feeding part **12** are integrally formed by using an independent sheet metal part, so that a structure is simplified, and integrity of an antenna is improved. The first feeding part **11** and the second feeding part **12** are located on two sides of the through hole **13**, so that isolation between parts near the first feeding part **11** and the second feeding part **12** can be improved, thereby improving radiation performance.

[0093] In a specific embodiment, the first feeding part **11** includes one first feeding bent section, and the first feeding bent section is connected to the first feeding structure **2** to generate a first

electromagnetic wave signal in a first polarization direction. The second feeding part **12** includes one second feeding bent section, and the second feeding bent section is connected to the second feeding structure **3** to generate a second electromagnetic wave signal in a second polarization direction.

[0094] Specifically, as shown in FIG. **6**, after the first feeding part **11** and the second feeding part **12** are stamped and formed, a first gap **14** and a second gap **15** that run through the radiator **1** are formed at a stamped position on the radiator **1**. Both the first gap **14** and the second gap **15** may be in a long strip shape. It may be understood that the first gap **14** and the second gap **15** in the long strip shape both have a long edge and a short edge. In this embodiment, the short edges of the first gap **14** and the second gap **15** are bent to form the first feeding part **11** and the second feeding part **12**, so that a part near the first feeding part **11** and a part near the second feeding part **12** approximately form two polarized orthogonal antennas.

[0095] The first feeding part **11** and the second feeding part **12** are parallel to each other, facilitating processing and manufacturing, and facilitating generation of a dual-polarized electromagnetic wave on the radiator **1**.

[0096] In another embodiment, as shown in FIG. **7**, the radiator **1** is stamped to form four gaps **16**, and the four gaps **16** are provided in a centrally symmetric manner around the through hole **13**. The first feeding part **11** includes two first feeding bent sections **111**, and the two first feeding bent sections **111** are respectively connected to ends of two gaps **16** that are provided in a centrally symmetric manner and that are close to the through hole **13** to generate a first electromagnetic wave signal in a first polarization direction. The second feeding part **12** includes two second feeding bent sections **121**, and the two second feeding bent sections **121** are respectively connected to ends of the other two gaps **16** that are provided in a centrally symmetric manner and that are close to the through hole **13** to generate a second electromagnetic wave signal in a second polarization direction. In this way, radiation performance of the antenna can be further improved.

[0097] In addition, in this embodiment, a dual-polarized electromagnetic wave needs to be generated on a same radiator **1**. In other words, a part of a region that is on the radiator **1** and that corresponds to the first feeding part **11** and a part of a region that is on the radiator **1** and that corresponds to the second feeding part **12** may approximately form two antennas whose polarization directions are mutually orthogonal. Currents between the two antennas are coupled, causing reverse cancellation of currents at some positions. Consequently, electric field distribution is uneven, and a stable directivity pattern cannot be obtained.

[0098] Therefore, as shown in FIG. **8** to FIG. **11**, this embodiment provides a first balancer **4** and a second balancer **5**, the through hole **13** is provided in a center of the radiator **1**, and the first balancer **4** and the second balancer **5** both pass through the through hole **13**. The first balancer **4** and the second balancer **5** may be of a same structure. The first balancer **4** includes a first coupling part **41** and a first ground part **42**. The first coupling part **41** and the first ground part **42** are respectively located on two sides of the radiator **1**. The first coupling part **41** is coupled to the radiator **1**. The second balancer **5** includes a second coupling part **51** and a second ground part **52**. The second coupling part **51** and the second ground part **52** are respectively located on the two sides of the radiator **1**. The second coupling part **51** is coupled to the radiator **1**.

[0099] The first coupling part **41** and the second coupling part **51** both keep a distance from the radiator **1**, to form a coupling connection. The radiator **1** may indirectly feed power to the first coupling part **41** and the second coupling part **51**. Currents on the first coupling part **41** and the second coupling part **51** can make an electric field of the radiator **1** more even, so that isolation between two parts whose polarization directions are different on the radiator **1** can meet a requirement on isolation between antennas. Therefore, a stable dual-polarized characteristic can be formed on one radiator **1**, and the dual-polarized antenna element can achieve a stable directivity pattern. In addition, a size of the antenna element can be reduced, miniaturization can be implemented, and space occupation can be reduced, thereby facilitating integration into a base

station antenna.

[0100] Specifically, as shown in FIG. 12, in a thickness direction of the radiator **1**, projection of the first coupling part **41** and projection of at least a part of the radiator **1** overlap, and projection of the second coupling part **51** and projection of at least a part of the radiator **1** overlap. In this way, the first coupling part **41** can separately correspond to parts that are on the radiator **1** and that are located on two sides of the first gap **14**, thereby improving an effect of current coupling between the first coupling part **41** and the radiator **1**, and helping make an electric field near the first gap **14** more even, to obtain a stable directivity pattern. Similarly, the second coupling part **51** also has a same effect as the first coupling part **41**, and details are not described herein again. Therefore, the first coupling part **41** and the second coupling part **51** can ensure overall evenness of an electric field of the dual-polarized antenna element, and ensure stability of the directivity pattern.

[0101] The first coupling part **41** and the second coupling part **51** may also be made of metal sheet metals, for example, thin sheet metal sheets. The first coupling part **41** and the second coupling part **51** may be in a shape such as a square, a rectangle, or a rhombus. In the dual-polarized antenna element shown in FIG. 9, the first coupling part **41** and the second coupling part **51** are rectangular. In the dual-polarized antenna element shown in FIG. 14, the first coupling part **41** and the second coupling part **51** are in a long strip shape.

[0102] Specifically, the first balancer **4** includes a first connection section **43**. The first connection section **43** passes through the through hole **13**. The first coupling part **41** and the first ground part **42** are respectively connected to two ends of the first connection section **43**. A bending angle between the first connection section **43** and the radiation surface **17** of the radiator **1** ranges from 80° to 100°.

[0103] As shown in FIG. 11, the first connection section **43** and a second connection section **53** may also be of structures such as a metal sheet, a metal column, a transmission line, and a microstrip. Certainly, for ease of processing and manufacturing, the first coupling part **41** and the first connection section **43** may be integrally formed by using a same material, and the second coupling part **51** and the second connection section **53** may be integrally formed by using a same material. The first connection section **43** and the second connection section **53** have a large length, and may run through the through hole **13** of the radiator **1**, so that the first coupling part **41** and the second coupling part **51** can be located on a same side of the radiator **1**, and the first ground part **42** and the second ground part **52** are located on another side of the radiator **1** and can be close to or directly connected to the ground, to form a closed loop of a circuit.

[0104] The radiator **1** radiates an electromagnetic wave to the outside by using the radiation surface **17** of the radiator **1**. It may be understood that a radiation direction of the electromagnetic wave is a direction perpendicular to the radiation surface **17**, and a direction of the electric field is perpendicular to the radiation direction of the electromagnetic wave, that is, is parallel to a direction of a surface of the radiator **1**. A current on the first connection section **43** and a current on the second connection section **53** flow along length directions of the first connection section **43** and the second connection section **53**. If the length directions of the first connection section **43** and the second connection section **53** are parallel to the radiator **1**, to be specific, if a flow direction of the current on the first connection section **43** and a flow direction of the current on the second connection section **53** are parallel to or have a small included angle with the direction of the electric field of the radiator **1**, the current on the radiator **1** is coupled to the current on the first connection section **43** and the current on the second connection section **53**. As a result, electric field and electromagnetic wave energy are weakened, and radiation performance of the antenna is reduced.

[0105] Therefore, in this embodiment, an angle between the first connection section **43** and the radiation surface **17** of the radiator **1** ranges from 80° to 100°. In this way, the first connection section **43** can be nearly perpendicular to the radiation surface **17** of the radiator **1**. In other words, a direction of the current on the first connection section **43** is nearly perpendicular to the direction

of the electric field of the radiator **1**, so that the current on the first connection section **43** can be prevented from being coupled with the current on the radiator **1**. Therefore, the antenna can achieve excellent radiation performance and the stable directivity pattern. Specifically, the angle between the first connection section **43** and the radiation surface **17** of the radiator **1** may be 80° , 85° , 90° , 95° , or 100° . More preferably, the first connection section **43** is perpendicular to the radiation surface **17** of the radiator **1**.

[0106] Specifically, the second balancer **5** includes the second connection section **53**. The second connection section **53** passes through the through hole **13**. The second coupling part **51** and the second ground part **52** are respectively connected to two ends of the second connection section **53**. A bending angle between the second connection section **53** and the radiation surface **17** of the radiator **1** ranges from 80° to 100° . Specifically, the angle between the second connection section **53** and the radiation surface **17** of the radiator **1** may be 80° , 85° , 90° , 95° , or 100° . More preferably, the second connection section **53** is perpendicular to the radiation surface **17** of the radiator **1**. A function and an arrangement effect of the second connection section **53** are the same as those of the first connection section **43**, and details are not described herein again.

[0107] A bending direction of the second coupling part **51** and a bending direction of the first coupling part **41** are opposite. After the first connection section **43** and the second connection section **53** pass through the through hole **13**, the first coupling part **41** may bend in a direction of the first gap **14**, and the second coupling part **51** may bend in a direction of the second gap **15**. The bending direction of the first coupling part **41** and the bending direction of the second coupling part **51** are away from each other, so that the first coupling part **41** and the second coupling part **51** can be separately coupled to the radiator **1** near the first gap **14** and the second gap **15**, thereby improving an electric field balance effect and helping obtain a stable directivity pattern.

[0108] Specifically, an angle between the first coupling part **41** and the radiation surface **17** of the radiator **1** and an angle between the second coupling part **51** and the radiation surface **17** of the radiator **1** range from 0° to 10° . In other words, the first coupling part **41** and the second coupling part **51** are separately nearly parallel to the radiation surface **17** of the radiator **1**. In this way, the first coupling part **41** and the second coupling part **51** can have large areas and can be coupled to the radiator **1**, thereby improving an effect of even distribution of the electric field. Specifically, the angle between the first coupling part **41** and the radiation surface **17** of the radiator **1** and the angle between the second coupling part **51** and the radiation surface **17** of the radiator **1** may be 0° , 2° , 4° , 6° , 8° , or 10° .

[0109] In this embodiment, the first coupling part **41** and the second coupling part **51** are disposed in parallel with the radiation surface **17** of the radiator **1** separately. In this way, a capability of evenly distributing the electric field of the radiator **1** by the first coupling part **41** and the second coupling part can reach a better level.

[0110] Specifically, as shown in FIG. **13**, a distance d between the first coupling part **41** and the radiation surface **17** of the radiator **1** and a distance d between the second coupling part **51** and the radiation surface **17** of the radiator **1** are less than 0.1 times a wavelength corresponding to a maximum operating frequency. In this way, a large bandwidth can be obtained, and it is ensured that the dual-polarized antenna element achieves good radiation performance and a stable directivity pattern.

[0111] Specifically, as shown in FIG. **13**, a maximum distance H between the first coupling part **41** and the first ground part **42** ranges from 0.25 times to 0.75 times a wavelength corresponding to a maximum operating frequency, and a maximum distance H between the second coupling part **51** and the second ground part **52** ranges from 0.25 times to 0.75 times the wavelength corresponding to the maximum operating frequency. In this way, the dual-polarized antenna element can achieve a constant directivity pattern, and has good radiation performance and isolation performance.

[0112] Specifically, an angle between the first feeding part **11** and the radiation surface **17** of the radiator **1** and an angle between the second feeding part **12** and the radiation surface **17** of the

radiator **1** range from 60° to 120° .

[0113] According to the foregoing descriptions, the radiator **1** radiates the electromagnetic wave by using the radiation surface **17** of the radiator **1**. It may be understood that the radiation direction of the electromagnetic wave is a direction perpendicular to the radiation surface **17**, and the direction of the electric field is perpendicular to the radiation direction of the electromagnetic wave, that is, is parallel to a direction of the surface of the radiator **1**. A current on the first feeding part **11** and a current on the second feeding part **12** flow along length directions of the first feeding part **11** and the second feeding part **12**. If the length directions of the first feeding part **11** and the second feeding part **12** are parallel to the radiator **1**, to be specific, if a flow direction of the current on the first feeding part **11** and a flow direction of the current on the second feeding part **12** are parallel to or have a small included angle with the direction of the electric field of the radiator **1**, the current on the first feeding part **11** and the current on the second feeding part **12** are coupled with the current on the radiator **1** before arriving at the radiator **1**. As a result, the electric field and electromagnetic wave energy are weakened, and the radiation performance of the antenna is reduced.

[0114] Therefore, in this embodiment, the angle between the first feeding part **11** and the radiation surface **17** of the radiator **1** and the angle between the second feeding part **12** and the radiation surface **17** of the radiator **1** range from 60° to 120° . In this way, the first feeding part **11** and the second feeding part **12** can be separately nearly perpendicular to the radiation surface **17** of the radiator **1**. In other words, a direction of the current on the first feeding part **11** and a direction of the current on the second feeding part **12** are nearly perpendicular to the direction of the electric field of the radiator **1**, so that the current on the first feeding part **11** and the current on the second feeding part **12** can be prevented from being coupled with the current on the radiator **1** before arriving at the radiator **1**. Therefore, the antenna can achieve excellent radiation performance and the stable directivity pattern. Specifically, the angle between the first feeding part **11** and the radiation surface **17** of the radiator **1** may be 60° , 70° , 80° , 90° , 100° , 110° or 120° . More preferably, the first feeding part **11** and the second feeding part **12** are separately perpendicular to the radiation surface **17** of the radiator **1**.

[0115] Specifically, as shown in FIG. **8**, the dual-polarized antenna element further includes a reflection panel **8**. The reflection panel **8** is located on a side that is of the radiator **1** and that is away from the radiation surface **17** of the radiator **1**. The first ground part **42** and the second ground part **52** are separately coupled to or directly connected to the reflection panel **8**.

[0116] The reflection panel **8** may be a metal sheet metal part, and may reflect an electromagnetic wave. It should be noted that two sides of the radiator **1** may radiate an electromagnetic wave. After the dual-polarized antenna element is integrated into the base station antenna, an electromagnetic wave radiated by a side that is of the radiator **1** and that faces the outside is a mainly received electromagnetic wave. A surface of this side of the radiator **1** may be defined as the radiation surface **17**. An electromagnetic wave radiated by the other side of the radiator **1** is blocked by an object, causing an energy loss.

[0117] In this embodiment, to reduce the energy loss, the reflection panel **8** may be disposed on a side that is of the radiator **1** and that is away from the radiation surface **17**. The reflection panel **8** can reflect an electromagnetic wave radiated by the surface that is of the radiator **1** and that is away from the radiation surface **17**, so that a radiation direction of the reflected electromagnetic wave is consistent with a radiation direction of the radiation surface **17**, thereby enhancing electromagnetic wave energy, improving signal reception sensitivity of the antenna, blocking and shielding interference from other electromagnetic waves from the back (a reverse direction) on a received signal, and improving antenna radiation performance.

[0118] The reflection panel **8** may be used as a ground of the antenna, and the first ground part **42** and the second ground part **52** may not contact the reflection panel **8** to implement coupled indirect grounding, or may contact the reflection panel **8** directly to implement direct grounding.

[0119] Specifically, as shown in FIG. 15, a plurality of dual-polarized antenna elements **10** may share one reflection panel **8**, and the plurality of dual-polarized antenna elements **10** form the antenna array **101**. In this way, integrity of the plurality of dual-polarized antenna elements **10** can be improved, and radiation performance of the antenna apparatus **100** having the plurality of dual-polarized antenna elements **10** can be improved.

[0120] Specifically, as shown in FIG. 11, the first ground part **42** and the second ground part **52** are connected to form an integrated structure. In this way, grounding consistency between the first balancer **4** and the second balancer **5** can be ensured. The first balancer **4** and the second balancer **5** may alternatively be designed as one part, facilitating production and manufacturing, and facilitating assembly of the balancer in the dual-polarized antenna element.

[0121] Specifically, as shown in FIG. 16 to FIG. 18, the dual-polarized antenna element further includes a third balancer **6** and a fourth balancer **7**. Structures of the third balancer **6** and the fourth balancer **7** may be the same as structures of the first balancer **4** and the second balancer **5**. The third balancer **6** and the fourth balancer **7** both pass through the through hole **13**. The third balancer **6** is centrally symmetrical to the first balancer **4**. The fourth balancer **7** is centrally symmetrical to the second balancer **5**. The third balancer **6** and the first balancer **4** correspond to one type of polarization, and the fourth balancer **7** and the second balancer **5** correspond to the other type of polarization. Through the provision of the four balancers, the dual-polarized antenna element can obtain a better polarization effect. As shown in FIG. 16, the first balancer **4**, the second balancer **5**, the third balancer **6**, and the fourth balancer **7** share a same ground part, in other words, the first balancer **4**, the second balancer **5**, the third balancer **6**, and the fourth balancer **7** form an integrated structure, and form an independent part, to facilitate producing and manufacturing, and assembly in the dual-polarized antenna element.

[0122] In an embodiment, as shown in FIG. 17, the radiator **1** has a first gap **14** and a second gap **15**. The first balancer **4** and the fourth balancer **7** correspond to the first gap **14**. The second balancer **5** and the third balancer **6** correspond to the second gap **15**. The first balancer **4** and the third balancer **6** correspond to one type of polarization, and the second balancer **5** and the fourth balancer **7** correspond to the other type of polarization.

[0123] In another embodiment, as shown in FIG. 18, there are four gaps **16** on the radiator **1**. The first balancer **4**, the second balancer **5**, the third balancer **6**, and the fourth balancer **7** each correspond to one gap **16**. The two balancers that are disposed in a centrally symmetric manner correspond to one type of polarization, so that an effect of balancing an electric field around the radiator **1** can be improved, and radiation performance of the antenna can be improved.

[0124] Specifically, the dual-polarized antenna element further includes a plastic support (not shown in the figure), the balancers, the radiator **1**, and the feeding part may be fastened on the plastic support. The plastic support may be installed on a structure such as the reflection panel **8**, to implement installation and fastening of the dual-polarized antenna element.

[0125] The foregoing descriptions are merely embodiments of this application, and are not intended to limit this application. For a person skilled in the art, this application may have various modifications and variations. Any modification, equivalent replacement, or improvement made without departing from the spirit and principle of this application shall fall within the protection scope of this application.

Claims

1. A dual-polarized antenna element, comprising: a radiator, wherein a first feeding part and a second feeding part of the radiator are integrally formed on the radiator, wherein the radiator has a through hole between the first feeding part and the second feeding part; a first feeding structure, coupled to, or directly connected to, the first feeding part; a second feeding structure, coupled to, or directly connected to, the second feeding part; a first balancer, wherein the first balancer passes

through the through hole, wherein the first balancer comprises a first coupling part and a first ground part, wherein the first coupling part and the first ground part are respectively located on two sides of the radiator, and wherein the first coupling part is coupled to the radiator; and a second balancer, wherein the second balancer passes through the through hole, wherein the second balancer comprises a second coupling part and a second ground part, wherein the second coupling part and the second ground part are respectively located on the two sides of the radiator, and wherein the second coupling part is coupled to the radiator.

2. The dual-polarized antenna element according to claim 1, wherein the first feeding part comprises a first feeding bent section, and wherein the first feeding bent section is connected to the first feeding structure and is configured to generate a first electromagnetic wave signal in a first polarization direction; and wherein the second feeding part comprises a second feeding bent section, and wherein the second feeding bent section is connected to the second feeding structure and is configured to generate a second electromagnetic wave signal in a second polarization direction.

3. The dual-polarized antenna element according to claim 1, wherein the first feeding part comprises at least two first feeding bent sections, and wherein the at least two first feeding bent sections are disposed in a centrally symmetric manner and are configured to generate a first electromagnetic wave signal in a first polarization direction; and wherein the second feeding part comprises at least two second feeding bent sections, and wherein the at least two second feeding bent sections are disposed in a centrally symmetric manner and are configured to generate a second electromagnetic wave signal in a second polarization direction.

4. The dual-polarized antenna element according to claim 1, wherein a projection of the first coupling part and projection of at least a part of the radiator overlap in a thickness direction of the radiator, and wherein a projection of the second coupling part and projection of at least a part of the radiator overlap.

5. The dual-polarized antenna element according to claim 1, wherein the first balancer further comprises a first connection section, wherein the first connection section passes through the through hole, wherein the first coupling part and the first ground part are respectively connected to two ends of the first connection section, and wherein an angle between the first connection section and a radiation surface of the radiator is between about 80° and about 100° .

6. The dual-polarized antenna element according to claim 5, wherein the second balancer further comprises a second connection section, wherein the second connection section passes through the through hole, wherein the second coupling part and the second ground part are respectively connected to two ends of the second connection section, wherein an angle between the second connection section and the radiation surface of the radiator is between about 80° and about 100° , and wherein a bending direction of the second coupling part and a bending direction of the first coupling part are opposite.

7. The dual-polarized antenna element according to claim 5, wherein an angle between the first coupling part and the radiation surface of the radiator and an angle between the second coupling part and the radiation surface of the radiator is between about 0° and about 10° .

8. The dual-polarized antenna element according to claim 7, wherein the first coupling part and the second coupling part are separately and each parallel to the radiation surface of the radiator.

9. The dual-polarized antenna element according to claim 1, wherein a distance between the first coupling part and a radiation surface of the radiator and a distance between the second coupling part and the radiation surface of the radiator are each less than 0.1 times a wavelength corresponding to a maximum operating frequency.

10. The dual-polarized antenna element according to claim 1, wherein a maximum distance between the first coupling part and the first ground part is between about 0.25 times and about 0.75 times a wavelength corresponding to a maximum operating frequency, and wherein a maximum distance between the second coupling part and the second ground part is between about 0.25 times

and about 0.75 times the wavelength corresponding to the maximum operating frequency.

11. The dual-polarized antenna element according to claim 1, wherein an angle between the first feeding part and a radiation surface of the radiator and an angle between the second feeding part and the radiation surface of the radiator are each between about 60° and about 120°.

12. The dual-polarized antenna element according to claim 1, wherein the dual-polarized antenna element further comprises a reflection panel, wherein the reflection panel is disposed on a side that is of the radiator and that is away from a radiation surface of the radiator, and wherein the first ground part and the second ground part are separately coupled to, or directly connected to, the reflection panel.

13. The dual-polarized antenna element according to claim 12, wherein the reflection panel is configured to be shared with at least one other dual-polarized antenna element.

14. The dual-polarized antenna element according to claim 12, wherein the first ground part and the second ground part are connected to form an integrated structure.

15. The dual-polarized antenna element according to claim 1, wherein the dual-polarized antenna element further comprises a third balancer and a fourth balancer, wherein structures of the third balancer and the fourth balancer are the same as structures of the first balancer and the second balancer, wherein the third balancer and the fourth balancer both pass through the through hole, wherein the third balancer is centrally symmetrical to the first balancer, and wherein the fourth balancer is centrally symmetrical to the second balancer.

16. The dual-polarized antenna element according to claim 1, wherein the radiator is a metal plate.

17. The dual-polarized antenna element according to claim 1, wherein the first feeding part and the second feeding part are integrally formed on the radiator and are stamped and bent elements.

18. An antenna apparatus, comprising: a dual-polarized antenna element, wherein the dual-polarized antenna element comprises: a radiator, wherein a first feeding part and a second feeding part are integrally formed on the radiator, wherein the radiator has a through hole, and wherein the through hole is located between the first feeding part and the second feeding part; a first feeding structure, coupled to, or directly connected to, the first feeding part; a second feeding structure, coupled to, or directly connected to, the second feeding part; a first balancer, wherein the first balancer passes through the through hole, wherein the first balancer comprises a first coupling part and a first ground part, wherein the first coupling part and the first ground part are respectively located on two sides of the radiator, and wherein the first coupling part is coupled to the radiator; and a second balancer, wherein the second balancer passes through the through hole, wherein the second balancer comprises a second coupling part and a second ground part, wherein the second coupling part and the second ground part are respectively located on the two sides of the radiator, and wherein the second coupling part is coupled to the radiator.

19. A base station system, comprising: an antenna apparatus, comprising: a dual-polarized antenna element, comprising: a radiator, wherein a first feeding part and a second feeding part are integrally formed on the radiator, wherein the radiator has a through hole, and wherein the through hole is located between the first feeding part and the second feeding part; a first feeding structure, coupled to, or directly connected to, the first feeding part; a second feeding structure, coupled to, or directly connected to, the second feeding part; a first balancer, wherein the first balancer passes through the through hole, wherein the first balancer comprises a first coupling part and a first ground part, wherein the first coupling part and the first ground part are respectively located on two sides of the radiator, and wherein the first coupling part is coupled to the radiator; and a second balancer, wherein the second balancer passes through the through hole, wherein the second balancer comprises a second coupling part and a second ground part, wherein the second coupling part and the second ground part are respectively located on the two sides of the radiator, and wherein the second coupling part is coupled to the radiator.

20. The base station system according to claim 19, wherein the first balancer further comprises a first connection section, wherein the first connection section passes through the through hole,

wherein the first coupling part and the first ground part are respectively connected to two ends of the first connection section, and wherein an angle between the first connection section and a radiation surface of the radiator is between about 80° and about 100°.
